

Phase 1: Mixed-Task Instruction Tuning



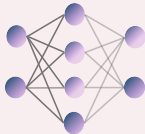
Detect if a given graph G contains any cycles.

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Assess if two nodes u and v in graph G are connected via a path.



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GraphWiz

$\rightarrow \mathcal{M}_{\text{sft}}$

Phase 2: DPO Alignment



Preferred Reasoning Path



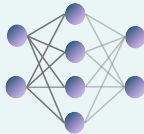
$\mathcal{R}_w$

$\vee$



$\mathcal{R}_l$

Maximum
Likelihood



GraphWiz

$\rightarrow \mathcal{M}$